

AlphaForge

A quantitative research stack — point-in-time event-driven backtesting, deflation-aware statistical evaluation, evolutionary multi-agent RL, and live paper-trading execution — applied honestly to a point-in-time S&P 500 universe (476 names, 2016–2025).

STATUS (2026-05-03): Tier 1 and Tier 2 both CLOSED FAILED on 2026-05-02. Zero strategies cleared the deflation-aware gauntlet. Project is in a pre-committed 30-day cooldown until 2026-06-01 before any substrate-change pivot.

Author: Atharva Patil

Components: JS frontend · Python alpha engine · MARL framework · Live paper-trading execution

Scale: 3 Python backend services · 541+ tests · 9 cross-sectional factors · 837 PIT S&P 500 membership events · 16 years of OHLCV

Companion artifacts: PHASE6_WRITEUP.md (Tier 1 final), TIER2_VERDICT.md (Tier 2 final), TIER1_STATUS.txt (master plan + outcomes)

Abstract

AlphaForge applies an end-to-end systematic-trading lifecycle — data, factors, event-driven backtesting, transaction-cost modeling, statistical evaluation, RL, and live execution — to nine cross-sectional equity signals and four linear combination strategies on a true point-in-time S&P 500 universe (476 tickers, 2,514 trading days, 2016–2025). Strategies and gate are pre-committed: DSR > 0.95 + bootstrap CI excludes zero + sign agreement, both of two non-overlapping OOS windows.

Tier 1 and Tier 2 both closed FAILED on 2026-05-02. 0 of 9 single factors and 0 of 4 Tier-1 combinations cleared. 0 of 8 Tier-2 lower-turnover strategies cleared. The closest result was a Markowitz overlay (MV) with alpha-residual OOS Sharpe +3.06 / +2.43 (CI [+1.83, +4.42] / [+1.39, +3.56]), alpha t-stats 4.33 / 3.43 (HC0), FF5+UMD R² 16% / 8% — statistically significant alpha that did not survive the 24-trial DSR deflation (0.92 / 0.70 vs. 0.95). Tier 2 falsified the row-2 "costs eat real signal" diagnosis: lowering rebalance horizon from 21d to 63d / 126d destroyed the alpha rather than preserving it, suggesting the MV-21 result is a short-horizon-specific phenomenon (most likely a 21-day residualized mean-reversion artifact) rather than a robust cross-sectional anomaly. The project's contribution is the methodology and infrastructure that produced these honest negative results — not surviving signals.

1	Executive Summary
2	Architecture & Data Pipeline
3	Alpha Engine & Event-Driven Backtest
4	Research Rigor & Cost Modeling
5	MARL Framework & Ablation Studies
6	Execution System & Live Risk Checks
7	Empirical Findings (Tier 1 + Tier 2)
8	Honest Limitations & Process Disclosures
9	Engineering Highlights
10	Verdict and the \$7 Cooldown

1. Executive Summary

Headline result. The pre-committed Tier 1 binary gate (DSR > 0.95 on FF5+UMD alpha-residual returns + bootstrap CI excludes zero + sign agreement, both OOS windows) FAILED. 0 of 9 single factors and 0 of 4 combination strategies cleared. The follow-up Tier 2 binary gate (lower-turnover variants of the row-2 diagnostic) ALSO FAILED: 0 of 8 strategies cleared and 0 near-misses. The project's research hypothesis — that some signal in the cross-sectional equity factor combination class survives a deflation-aware gauntlet on this substrate — is rejected at both tested levels.

The interesting failure. A Markowitz overlay over the 9 per-factor long-short return series produces alpha-residual OOS Sharpe +3.06 / +2.43 with alpha t-stats 4.33 / 3.43 (HC0 SEs) and FF5+UMD R² of 16% / 8%. The signal is genuinely orthogonal to the standard factor model and statistically significant under any conventional bar ($p < 1e-4$ / $p < 1e-3$). It fails the gate *only* on the DSR hurdle (0.92 / 0.70 vs. the pre-committed 0.95) when deflated against the 24-trial set. Tier 2 then falsified the row-2 interpretation: lowering rebalance from 21d to 63d / 126d destroyed the alpha (3.06 → 0.79 → 0.95 in OOS-A), suggesting the MV-21 result is a 21-day-specific residualized reversal phenomenon rather than a real cross-sectional anomaly trapped behind a cost wall.

What the project actually contributes. The methodology and infrastructure that produced the result above. AlphaForge implements the full systematic-trading lifecycle — data ingestion, factor construction, point-in-time event-driven backtesting, transaction-cost modeling, deflation-aware statistical evaluation, post-portfolio FF5+UMD alpha residualization, neuroevolutionary multi-agent RL, and a live paper-trading execution loop with kill-switch risk logic. The statistical layer integrates the Deflated Sharpe Ratio (Bailey & López de Prado), Hansen's Superior Predictive Ability test, White's Reality Check, López de Prado's Purged K-Fold cross-validation, and stationary-bootstrap confidence intervals on both raw long-short returns and FF5+UMD alpha-residual returns.

One process disclosure deserves headline-level visibility. While running the Tier 2 diagnostic, a load-bearing methodology bug was discovered in the residualization wiring: `prepare_analysis_returns()` was returning raw returns regardless of the residualize flag, and the `compute_portfolio_alpha` post-hoc layer was not wired into the main gauntlet. The intended residualization layer existed as dead code for an unknown period; the JSON metadata claimed `analysis_returns_mode: residualized` while the actual computation was on raw returns. Fixed in ~50 lines (pass `reference_factors` into `_run_variant`, populate per-OOS-window `ff5_alpha` blocks, both gates auto-detect and use alpha-residual Sharpes when available). All headline numbers in this document are post-fix. Pre-fix outputs are preserved as `*_residualized.json` backups for full auditability.

Platform Metrics at a Glance

- **541+ tests** across the three Python subprojects (core + MARL + Execution); all green post-bug-fix.
- **9 cross-sectional alpha factors** (Momentum 12-1, Mean Reversion 5d, Volume Surge, RSI Divergence, Earnings Drift, Amihud Illiquidity, Idiosyncratic Volatility, Residual Reversal 5d, Low Volatility) plus 4 pre-committed linear combinations (EWE, ICW, MV, ICW-flip).

- **Point-in-Time S&P 500 Universe** with 837 chronological membership events 2010–2026 (407 REMOVE + 352 ADD + 78 RENAME), built from Wikipedia revisions + EDGAR CIK enrichment, validated to 99% monthly correlation against ^SP500EW.
- **Event-driven simulation engine** enforcing strict no-look-ahead (BarHistory raises if queried past as_of) and per-fill commission/slippage accounting; previous flat-bps engine (real_engine.py) retired in Phase 2 consolidation.
- **FF5+UMD alpha-residualization layer** (post-portfolio time-series regression via compute_portfolio_alpha) wired into both Phase 4 single-factor and Phase 5 combination evaluation, with HC0 heteroskedasticity-consistent SEs on the alpha intercept.

2. Architecture & Data Pipeline

The project is partitioned into four primary components:

Component	Focus Area
JS Frontend	Glassmorphic terminal UI, interactive Chart.js visualizations, slide-out ticker panels. Strict numerical formatting.
alphaforge-python	Data store, 9-factor engine, event-driven backtest, optimization, FF5+UMD residualizer, gauntlet.
alphaforge-marl	NSGA-II evolution + PPO + FOMAML agent training and HMM regime detection. Existing checkpoints.
alphaforge-execution	Live paper trading (yfinance/Alpaca), kill-switch risk logic, and SQLite persistence. .halt engaged.

Point-in-Time Universe Substrate

Phase 1 of Tier 1 replaced the original 50-name today-surviving universe with a true PIT membership log built from Wikipedia revision history + SEC EDGAR CIK enrichment. The construction parses 2,811 page revisions, applies a hybrid byte-delta + comment-keyword pre-filter, normalizes share-class punctuation (↔) before CIK lookup, and runs an action-precedence + suspect-pair guard on the differ. Output: 837 events (407 REMOVE + 352 ADD + 78 RENAME) verified by 12/12 hand-built spot-check fixtures, cross-checked at 84% against Wikipedia's curated changes table, and reconciled at 0.9895 monthly return correlation against ^SP500EW. Canonical accessor: data.market.pit.validator.membership_on_date.

Unidirectional Data Flow

Historical OHLCV originates from yfinance via a single sync script (sync_market_data.py) and is materialized as per-ticker-per-year Parquet files in data/quarantine/market/. The Parquet store is the immutable source of truth; the alpha engine, MARL trainers, and research studies all consume this local dataset via cached loaders, ensuring training and evaluation never hit the network and are perfectly reproducible. Known data gap: 226 of 881 PIT ever-member tickers have no yfinance OHLCV (delisted / restructured); flagged honestly in every downstream metric.

3. Alpha Engine & Event-Driven Backtest

The core Python package features a vectorized factor construction framework and an event-driven backtest engine designed to replicate live trading mechanics.

Cross-Sectional Factors

Nine factors are evaluated in the gauntlet (5 matching the JS frontend for parity, 4 Python-only):

Factor	Brief Description
Momentum (12-1)	Trailing 12-month return excluding the most recent month.
Mean Reversion (5d)	Inverse of 5-day return.
Volume Surge	Short-term vs long-term volume moving average.
RSI Divergence	Standard 14-day Relative Strength Index minus 50.
Earnings Drift	Post-earnings announcement drift proxy (10-day return).
Amihud Illiquidity	Absolute return divided by dollar volume.
Idiosyncratic Volatility	Negated annualized residual vol vs equal-weight market (60d window).
Residual Reversal (5d)	Negated 5-day sum of residuals against the equal-weight market.
Low Volatility	Negated annualized 60d log-return stdev (Ang/Hodrick/Xing/Zhang 2006).

Event-Driven Engine

Replaces the legacy vectorized panel sweep to strictly enforce causality and realistic trade execution. Phase 2 engine consolidation is complete:

- **BarHistory**: Point-in-time data structure that raises errors if queried past its current `as_of` date.
- **ExecutionHandler**: Requires next-bar timestamps for order fills and enforces flat-slippage/commission models directly on `FillEvents` (per-fill, not flat post-hoc deduction).
- **Portfolio**: Tracks cash, positions, and marks-to-market. Fails loudly on missing price data.
- **Retired**: `real_engine.py` was removed in Phase 2 because its same-bar fills, daily clamp, and flat rebalance-cost deduction were architecturally wrong. Regression test `test_engine_consolidation.py` gates the deletion.

4. Research Rigor & Cost Modeling

AlphaForge produces standardized, reproducible research reports using the same gauntlet kernel across Tier 1 single-factor, Tier 1 combination, and Tier 2 lower-turnover evaluations:

- **Information Coefficient + decay** across {1, 5, 10, 21, 63}-day horizons (Spearman, t-stat, IR, hit-rate).
- **Quintile-spread long-short backtest** with monthly rebalance (or 63d / 126d in Tier 2), cost-net of commission + half-spread + linear impact ($10\text{bp} \times \text{turnover}$).
- **Stationary-bootstrap Sharpe CI** (Politis & Romano 1994); Tier 1 used 2,000 reps with 21d mean block, Tier 2 used 4,000.

- **Hansen SPA** (2005) and **White's Reality Check** (2000) across the full trial matrix per OOS window.
- **Deflated Sharpe Ratio** (Bailey & López de Prado 2014), deflated against the full pre-committed trial set.
- **Purged-embargoed K-fold CV IC** (López de Prado 2018) at $h=21$.
- **Post-portfolio FF5+UMD alpha residualization** via `compute_portfolio_alpha`: time-series regression of each strategy's daily returns on the six-factor reference series, with HC0 SEs on the alpha intercept and bootstrap CI on the residual Sharpe.

Cost Model — Honest Documentation of an Underestimate

`cost_model.py` contains both the parametric model used in Tier 1 / Tier 2 (1bp commission + 2bp half-spread + 10bp/turnover linear impact) and the Corwin-Schultz half-spread estimator (Corwin & Schultz 2012). The Tier 2 Phase 1 sanity check found that the Corwin-Schultz median half-spread is **7-8 bps across all windows**, vs. the parametric model's 2 bps — meaning Tier 1 substantially underestimated transaction costs by $\sim 3\text{-}4\times$. Per pre-commit, Tier 2 did not recalibrate parameters mid-evaluation; the divergence is documented in `tier2_phase1_cost_check.json` as a known limitation. Implication: the Tier 2 "lower-turnover helps" hypothesis should have been MORE supported under realistic costs, not less. It still failed.

5. MARL Framework & Rigor Report

The `alphaforge-marl` subsystem trains a population of multi-agent reinforcement-learning policies via neuroevolution + PPO + first-order MAML, with a regime-conditional Thompson-sampling allocator at the top. The ambition was end-to-end: signal extraction, position sizing, and capital allocation as learned components.

- **Evolutionary PPO + MAML**: Population-based training via NSGA-II selection on Sharpe, drawdown, turnover.
- **Regime Bandit**: HMM (Baum-Welch EM) detects regimes; Thompson sampling allocates capital among ensemble agents.
- **Walk-Forward Validator**: Anchored splits with strict temporal isolation.
- **Ablation Ladder**: Paired stationary-bootstrap Sharpe-difference tests across configurations.

MARL Honest Verdict ([research/marl_rigor_report.md](#))

The MARL rigor report enumerates 100+ generation-level trials and applies the same statistical hygiene as the single-factor study. The verdict: **0% of trials beat the equal-weight baseline on the same window**. Mean baseline-excess Sharpe is -1.13 ; the best individual trial's baseline-excess is -0.669 (still negative). Honest OOS DSR (mean stability Sharpe across 2 seeds, deflated against the 100-trial set) is 0.038 vs. the 0.95 hurdle. The report's conclusion: *the agents learned beta to the equal-weight basket, not alpha*. Per Tier 1 + Tier 2 not-doing lists, no further MARL training is in scope.

6. Execution System & Live Risk Checks

The alphaforge-execution subsystem implements the daily live trading loop against Alpaca. **Currently halted: .halt is engaged.** The 10 paper positions across momentum and MARL accounts were flattened on 2026-04-26. Re-arm requires the four conditions in TIER1_PAUSE.md (Tier 1 gate passed, signal is the survivor, universe expanded, ≥ 6 months paper trade). With Tier 1 + Tier 2 failed, those conditions cannot be met from current state; the .halt stays on indefinitely.

Kill Switch & Unwind Ladder

The KillSwitch enforces strict constraints from execution_config.yaml. It monitors 6 triggers: max drawdown, single-day loss, consecutive losing days, realized slippage median, realized cumulative fill-error, and minimum liquid ticker counts. If tripped, it blocks new entries and executes a 3-stage unwind ladder (25% immediate, 50% at +4h, 100% by next close), requiring manual acknowledgment to re-arm.

Slippage Reconciliation

slippage_reconciliation.py compares executed trades in the SQLite database against the backtest's assumed slippage. Computes KS tests and cumulative NAV drag to detect execution decay. Currently limited to 7 lifetime fills (pre-halt); not enough for a load-bearing tracking-error number.

7. Empirical Findings (Tier 1 + Tier 2)

The full statistical gauntlet was applied to all nine single factors, all four Tier-1 combinations, and all eight Tier-2 lower-turnover strategies. Results below are post-bug-fix (FF5+UMD alpha-residual where the residualization layer applies). No factor or strategy is filtered out for narrative reasons.

Tier 1 — Single-Factor Gauntlet (alpha-residual)

Factor	OOS-A α -SR	OOS-B α -SR	Verdict
Momentum (12-1)	-1.48	-0.94	fail
Mean Reversion (5d)	-0.45	+0.35	fail
Volume Surge	-1.55	-1.28	fail
RSI Divergence	-1.78	-1.15	fail
Earnings Drift	-1.32	-1.24	fail
Amihud Illiquidity	+0.19	-0.53	fail
Idiosyncratic Volatility	-0.45	-0.17	fail
Residual Reversal (5d)	-1.83	-1.63	fail
Low Volatility	-0.91	-0.44	fail

Survivors: 0 of 9. Most factors land negative in at least one window; several are negative in both. Bootstrap CI excludes zero on the wrong side for 6 of 9; sign disagreement across windows for 2 of 9.

Tier 1 — Combination Gauntlet (alpha-residual)

Strategy	OOS-A α -SR	OOS-B α -SR	DSR-A	DSR-B
EWE	-1.71	-0.94	0.000	0.000
ICW	-0.60	-1.00	0.000	0.000
MV	+3.06	+2.43	0.920	0.701
ICW-flip	-0.60	-1.00	0.000	0.000

Survivors: 0 of 4. MV is the lone non-trivial result: alpha t-stats 4.33 / 3.43 (HC0), FF5+UMD R² 16% / 8%, bootstrap p_positive = 1.0 in both windows. Failed only on the DSR hurdle when deflated against the 24-trial set.

Tier 2 — Lower-Turnover Gauntlet (alpha-residual)

Strategy	OOS-A α -SR	OOS-B α -SR
MV-63	+0.79	+1.97
MV-126	+0.95	+0.11
MV-63-volcap	+0.79	+1.97
MV-126-volcap	+0.95	+0.11
MV-63-shrunk	+0.62	+1.85
MV-126-shrunk	+0.87	-0.31
MV-63-ext	+0.79	+1.97
MV-126-ext	+0.95	+0.11

Survivors: 0 of 8. Near-misses (α -SR $\geq +1.5$ in BOTH windows): 0. Pre-committed \$5.2 outcome 3: clean fail. The R1k Tier 2.5 contingent is NOT activated.

The headline non-obvious finding: the MV-21 baseline alpha of +3.06 / +2.43 collapses to +0.79 / +1.97 at quarterly rebalance and +0.95 / +0.11 at semi-annual rebalance. Lower turnover *destroyed* the alpha. This is the inverse of what the row-2 "costs eat real signal" hypothesis predicted, and it falsifies the Phase 6 §4 diagnostic. The MV-21 alpha is most likely a 21-day-specific residualized mean-reversion artifact (Da/Liu/Schaumburg 2014) rather than a robust cross-sectional anomaly.

MARL Rigor (existing checkpoints, no new training)

Metric	Value
Total trials enumerated	100
OOS mean stability Sharpe (2 seeds, 251d)	+0.72

OOS stability DSR (deflated for trials)	0.038
Reward-mix trials beating equal-weight	0 / 60
Mean baseline-excess Sharpe	-1.13

Reading: agents learned beta to equal-weight, not alpha. Adding more training would not address the underlying problem; the MARL framework may be redirectable to support roles (execution, sizing, regime-routing) in a future Tier 3, but as an alpha source it does not survive the gauntlet. Detailed report: [alphaforge-marl/research/out/marl_rigor_report.md](#).

Aggregate interpretation. Across 9 single factors + 4 Tier-1 combinations + 8 Tier-2 strategies + a multi-agent RL ensemble, evaluated on a true PIT S&P 500 substrate with FF5+UMD post-portfolio alpha residualization and a 24-trial DSR deflation, no construction in the tested space produces a deflation-survivable, sign-consistent signal. The closest result was suggestive but not survivable; the falsification test (lower turnover) demonstrated the suggestive result was construction-specific rather than fundamental. None of the implemented signals or ensembles is fit to be allocated capital against.

8. Honest Limitations & Process Disclosures

These are limitations of the *current* work and disclosures about how the work was conducted. The framework's methodology is sound; the items below describe specific gaps, choices, and process failures.

- **Residualization-wiring bug (process failure, fixed 2026-05-02).** `prepare_analysis_returns()` returned raw returns regardless of the residualize flag for an unknown period. The post-hoc `compute_portfolio_alpha` layer existed but was not wired into the main gauntlet (`_run_variant` was called without `reference_factors`). JSON metadata claimed `analysis_returns_mode: residualized` while the actual computation was on raw returns. Caught during the Tier 2 diagnostic; fixed in ~50 lines; gauntlet re-run; verdict held but the documented diagnostic shifted. An audited institutional pipeline would have caught this via code review or independent reimplemention.
- **Phase 3 FF5 replica gate was soft-passed.** 3 of 6 reference factors (SMB, RMW, CMA) failed the >0.85 correlation threshold against Ken French's published series on this 500-ticker substrate (structural — French builds on full CRSP). The decision was to use French's published series for residualization rather than the local replica, sidestepping the failed sub-gate. This is documented but is a partial compromise of the pre-commitment.
- **Tier 2 design contained two no-op variants.** Vol-cap variants (MV-{63,126}-volcap) are mathematical no-ops because Sharpe and DSR are scale-invariant. Extended-history variants (MV-{63,126}-ext) didn't actually use extended training data given the panel start = 2016, so `train_start=2010` sliced the same data as `train_start=2016`. The trial set effectively reduced from 8 to 4 unique strategies. Verdict held under both interpretations.
- **Cost model under-estimates real spreads.** Parametric 2bp half-spread vs. Corwin-Schultz median 7-8bp across all windows. Tier 1 / Tier 2 results assume costs ~3-4× lower than realistic. Effect: makes the row-2 "lower turnover helps" hypothesis look *more* viable than it would under realistic costs; the verdict's robustness is strengthened, not weakened, by the underestimate.

- **25% data gap on the PIT universe.** 226 of 881 ever-member tickers have no yfinance OHLCV (delisted / restructured). Documented in every metric. CRSP-grade data would close this gap but was out of budget by the founder-path discipline.
- **No live-vs-backtest tracking number.** 7 lifetime fills before the .halt; not enough for KS-test or cumulative-drag analysis. The infrastructure exists; the data does not.
- **No per-name borrow-cost differentiation.** Borrow-cost table supports HTB overrides; currently populated with general-collateral defaults. Material understatement for any non-mega-cap short leg.
- **Trial count is conservative for MARL.** The MARL DSR deflates against 100 generation-level trials; the true search space (architecture, curriculum, reward shaping, selection rule) is larger. The published OOS DSR is an optimistic *upper* bound on credibility.

9. Engineering Highlights

- **JS / Python numerical parity to 10 decimal places** on the PRNG, factor scoring, and backtest paths, enforced by parity-fixture tests. The same research expressible in either runtime without numerical drift.
- **Defensive numerics by construction.** A small set of primitives — `safe_div`, `sanitize_number`, `clamp`, `validate_series` — is used uniformly across both runtimes; NaN / Inf cannot propagate through the factor pipeline.
- **Architectural enforcement of no-look-ahead.** BarHistory raises if asked for any row past its `as_of`; ExecutionHandler rejects fills that aren't strictly later than their originating order. PIT enforced by the type system, not by reviewer vigilance.
- **Pre- and post-fix output preservation.** The pre-bug-fix Tier-1 outputs are kept as `*_residualized.json` backups alongside the post-fix outputs, enabling full audit of the diagnostic shift between the two runs.
- **One-command reproducibility.** make all rebuilds every research artifact in this document — factor study + Phase 5 combination + Tier 2 gauntlet + both gates — from the parquet store in ~5 minutes.
- **CI drift detection on headline metrics.** A GitHub Actions matrix runs the full test suite and re-runs each headline study, diffing rebuilt JSON against the committed artifact. A silent numerical regression in any factor or metric fails the build.

10. Verdict and the §7 Cooldown

Tier 1 verdict (2026-05-02): the pre-committed binary gate FAILED. 0/9 single factors, 0/4 combinations cleared. MV nearly survived; killed by deflation. Phase 6 (the writeup) committed the diagnostic to row 2 of the failure-path matrix ("real signal eaten by costs/multiple-testing"). Full writeup: PHASE6_WRITEUP.md.

Tier 2 verdict (2026-05-02, same day): the row-2 hypothesis was tested with 8 pre-committed lower-turnover strategies and FAILED decisively. 0 strategies cleared, 0 near-misses. Lower turnover destroyed the alpha rather than preserving it, falsifying the row-2 diagnosis. The revised reading is

that the MV-21 alpha is most likely a 21-day-specific residualized reversal artifact, not a robust cross-sectional anomaly. Full writeup: TIER2_VERDICT.md.

Current state — §7 reset cooldown until 2026-06-01. Per the pre-committed Tier 2 design §7, no new gauntlet design, no Tier 3 design, no MARL revival, no live re-arming, and no paid-data subscriptions are permitted in this 30-day window. The substrate-change reassessment memo is the single AlphaForge artifact that may land before 2026-06-01; it asks whether the cross-sectional equity factor + linear combination + parametric cost construction class is the right substrate at all, or whether the next research arc should pivot to futures, market-making, options, crypto, or away from systematic alpha entirely. That memo unblocks the next decision.

What the project demonstrates today. An end-to-end research-grade systematic-trading stack built and run on free data by a solo undergraduate, applied honestly to a known-hard problem, with the methodology bug found and fixed in the same session it surfaced and the resulting diagnostic shift documented openly. The negative results published here are the artifact; surviving signals are not.

Repository: *Quant Alpha* (AlphaForge). Generated by docs/build_alphaforge_pdf.py. Reproducible output; JSON data artifacts under research/out/ and research/out/tier2/ drive the specific empirical numbers shown.